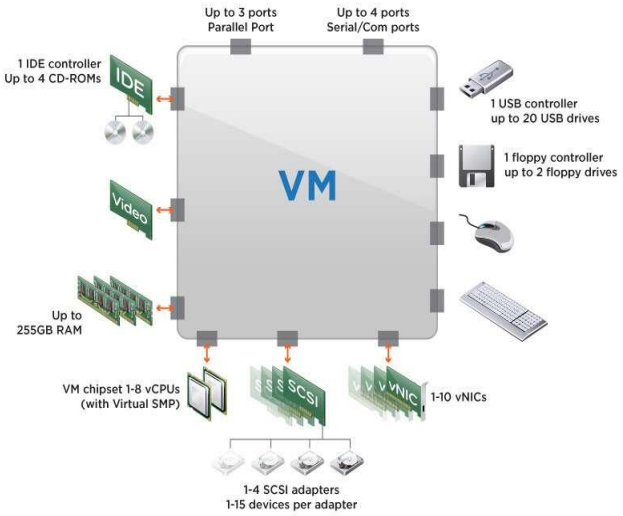


CLOUD COMPUTING COURSE

Lecture28- Virtualization Features(3)

ABSTRACT

Lecture28 Introduces VM features.

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Virtualization Features

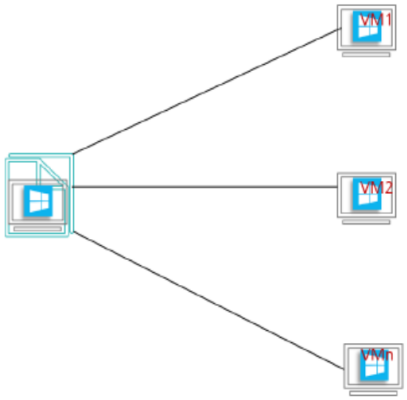
VM Features

1. Quick VM Deployment

When we purchase an Elastic Cloud Server (ECS) on the public cloud, a VM with an OS will be generated in the background. The time required for installing an OS on the VM is **much shorter** than that required for installing an OS on a PC. **This is the quick VM deployment feature.**

Methods to deploying VMs quickly: VM can be deployed by using:

1. **VM Templates**
2. **Cloning other VMs.**



1. VM Template:

- A VM template is a copy of a VM. It contains VM disks and VM configuration files.
- Creating a VM using a VM template can greatly **reduce the time for configuring** a new VM and installing an OS.

- After a VM template is created, it **cannot be started or modified**. This ensures that the template will not be modified by other users and does not occupy compute resources of the cluster.

- A VM deployed using a template **is independent** of the template.

- To update or edit the template, you need to convert the template to a VM first. After the template is edited, you can convert the VM into a template.

- **Scenarios**: VM templates are useful for deploying a large number of similar VMs because it can keep **VM consistency** and **parameters can be modified automatically (such as the host name, IP address)** as required.

- For example, if a group of test personnel needs to test the newly developed software, the administrator can use a VM with the software installed as a template, quickly deploy a batch of VMs with the same configuration, and assign the VMs to different test personnel for testing in different scenarios and with different requirements. If any problem occurs during the test, the administrator can delete the faulty VM and deploy a new VM for the test personnel.

- In addition, different VM templates may include different software. For example, a financial system may be pre-installed in a VM template used by financial personnel, and an ERP system may be pre-installed in a VM template used by sales personnel. VMs can be created for different departments using different templates.

2. VM Cloning: In addition to deploying a VM using a template, you can also **quickly deploy a VM using another VM**. This function is called **VM cloning.**

- Different from template-based deployment, VM cloning is a process of completely copying the source VM at a specific time point. All settings of each cloned VM, including the host name and IP address, are the same as those of the source VM. The system reports an error if there are two identical IP addresses in a LAN. Therefore, the cloned VM and the source VM cannot be started at the same time.

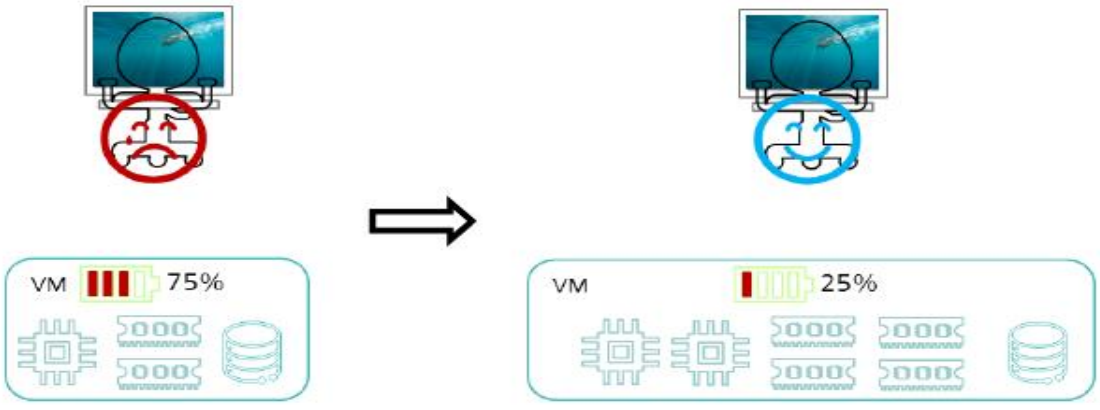
3. VM Resource Hot-Add

Hot-add is to add compute, storage, and network resources to a VM when the VM is started.
From the administrator's perspective, the CPU and memory parameters of a VM are part of the VM configuration file. The hardware configuration of the VM can be modified through modification of these parameters.

In addition to CPU and memory resources, storage and network resources also support hot-add. For example, the disk capacity of a VM can be expanded or a NIC can be added to a VM.

The hot-add function needs to be supported by the VM and OS so that the added resources can take effect immediately. Otherwise, the VM must be restarted and can be used only after the OS identifies the hardware resources.

In most cases, resources **support hot-add but do not support hot-delete**. For example, the administrator can expand the capacity of a disk from 10 GB to20 GB but cannot necessarily reduce the disk capacity from 20 GB to 10 GB.
In terms of storage resource addition, you can **expand the capacity** of **existing** disks and **add new disks to VMs**.



VM resource hot-add